

# Calibrated Joint Uncertainty Across Scales for Small-Area Estimation: A Bayesian Model-Based Geostatistics Framework with a Semi-Dense Coarse-Layer Tree Prior

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## Abstract

Small-area estimation (SAE) from complex survey data must carry honest uncertainty from the point/cluster scale at which modeling occurs to the area scale at which policy decisions are made. We present a Bayesian model-based-geostatistics (MBG) framework that couples a Bayesian additive regression tree (BART) mean, a structured SPDE–Matérn Gaussian-process (GP) spatial field, and Pólya–Gamma augmentation for the binomial-logit likelihood, and that produces genuine *joint* posterior draws of the latent prevalence surface. From these draws, credible intervals for any area-average functional are obtained by aggregation, so the same fitted model delivers uncertainty quantification (UQ) at the pixel, cluster, and administrative-area scales simultaneously. Across a ten-method benchmark, this framework is the only one whose calibration ratio (root-mean-square error over posterior standard deviation) stays near one at *every* aggregation scale; the competitors are each calibrated at only a single crossing scale. The framework has one residual soft spot: the area (Admin-1) coverage sits modestly below nominal because the flexible tree mean carries a slowly-vanishing low-frequency (“non-Donsker”) bias that survives aggregation. We diagnose this as a *center* bias—not a width problem—and remedy it with a *semi-dense coarse-layer tree prior*: a hard depth-floor  $d_0 \approx \sqrt{\log N}$  that forces the coarsest tree layers to remain dense, after [Castillo and Ročková \(2021\)](#). The depth-floor is a single prior knob (three localized edits, essentially zero added compute) that is *exactly* the hypothesis our target forward-route functional Bernstein–von Mises (BvM) theorem requires. In a paired, matched-seed simulation study the floor recenters Admin-1 80% coverage toward nominal across a broad plateau of  $d_0 \geq 1$  ( $0.79 \rightarrow 0.815\text{--}0.827$ ) and across aggregation scales and spatial amplitudes, *without* disturbing the near-nominal pixel/cluster/95% calibration. We are deliberate about the strength of evidence: the primary paired contrast is right-signed but modest ( $\Delta\text{cov}_{80} = +0.014 \pm 0.010$  at 24 replicates, below two Monte-Carlo standard errors), and the corroborating sensitivity, scale, and adversarial sweeps—not the single primary contrast—carry the claim. On the theory side we state a Matérn-only Admin-1 functional BvM (**[PROVEN]**, modulo a re-regime fix) and a single-tree Gaussian combined BvM (**[CONJECTURED]**, no fatal holes), and we are explicit that the binomial-logit transfer and the deployed-dimension ( $p \approx 49$ ) regime remain open.

**Keywords:** small-area estimation; model-based geostatistics; Bayesian additive regression trees; Pólya–Gamma; Matérn/SPDE; functional Bernstein–von Mises; calibrated uncertainty; survey statistics.

## 1 Introduction

National health and development programs are increasingly governed by *small-area* statistics: the prevalence of child wasting in an administrative district, the under-five mortality rate in a sub-

county, the coverage of an intervention in a province. These quantities are estimated from complex household surveys—most prominently the Demographic and Health Surveys (DHS; [Kenya National Bureau of Statistics and ICF International, 2015](#))—that are designed to be representative at coarse scales but are sparse, clustered, and stratified at fine scales. The statistical challenge is a *scale mismatch*: the policy deliverable is an area-level average, but the data and the natural model live at the point/cluster level. A credible method must therefore carry honest uncertainty *through aggregation*, from the clusters where measurements are taken to whatever administrative partition a decision-maker cares about, and must do so when the covariate–response relationship is nonlinear and there is residual spatial structure the covariates do not explain.

**Two cultures, each calibrated at one scale.** Existing methodology splits into two cultures, and—this is the gap we target—each is calibrated at only one aggregation scale. The *design-based* culture (direct/Hájek estimators, model-assisted generalized regression, and the more recent prediction-powered inference of [Angelopoulos et al., 2023](#)) is correct where its independent-residual variance model holds, but it supplies no pixel- or cluster-level uncertainty and understates standard errors under design effects. The *model-based geostatistics* (MBG) culture ([Diggle et al., 1998](#); [Rue et al., 2009](#); [Lindgren et al., 2011](#)) places a latent Gaussian field over space and aggregates it to areas; the INLA SPDE–Matérn model is the standard in low- and middle-income-country (LMIC) disease mapping, and disaggregation regression ([Nandi et al., 2023](#)) is its native binomial, joint-UQ analogue. Both rest on a *linear* mean. Where the true mean is curved or interacting, that linear mean misfits, and the misfit surfaces as a *scale-dependent* coverage error: calibrated at the coarse crossing scale, miscalibrated elsewhere. Flexible-mean machine learning enters from two directions but does not close the gap. Bayesian additive regression trees (BART; [Chipman et al., 2010](#)) give a well-calibrated *pointwise* nonparametric mean, but with no spatial term they collapse at the area functional (we measure Admin-1 80% coverage of 0.29 for vanilla BART). Spatial random forests (RF-GLS; [Saha et al., 2023](#)) and related spatial-ML methods restore a spatial mean, but emit only *marginal* per-cluster intervals and reach nominal coverage through a constant bias or a laundered interval width that holds at one scale and fails at another. None of these delivers a coherent *joint* area-mean law that is calibrated across scales.

**Contribution.** We make the calibrated-joint-UQ-across-scales property the headline and deliver it with a framework—BARTSIMP-PG-SD—built from three known ingredients and one new prior knob. The mean is a BART ensemble; the spatial field is an SPDE–Matérn GP ([Lindgren et al., 2011](#)); the binomial-logit likelihood is handled by Pólya–Gamma augmentation ([Polson et al., 2013](#)), which renders the model conditionally Gaussian and lets us draw the latent surface *jointly* across all clusters. From those joint draws, the posterior of any bounded linear functional of the surface—in particular the area-average prevalence—follows by aggregation, carrying the full cross-cluster covariance that marginal-interval competitors lack. On a ten-method benchmark this framework is the unique method whose calibration ratio stays near one at every aggregation scale (Section 5).

The framework has one residual soft spot, and identifying and fixing it is the specific methodological novelty. The Admin-1 coverage sits modestly *below* nominal ( $\text{cov}_{80} \approx 0.77$  versus 0.80) while the 95% coverage is near-nominal and the calibration ratio is only slightly above one (validation ratio  $\approx 1.034$ ; production-surface ratio  $\approx 1.08$ ). Because the ratio exceeds one only slightly while the lower quantile lags, the deficit is a *center* bias, not a width problem: a standardized bias of  $\delta = \sqrt{\text{ratio}^2 - 1}$ , i.e.  $\delta \approx 0.26$  at the validation ratio and  $\delta \approx 0.4$  posterior SD at the production-surface ratio (Section 3.4). This is the predicted *non-Donsker* effect of the tree mean: the flexible mean carries a slowly vanishing low-frequency bias that the credible band cannot see,

and that survives area-averaging precisely because the area mean loads on the coarse, low-resolution component of the surface—the very component a pruning tree prior under-represents. [Castillo and Ročková \(2021\)](#) prove, in a single-tree Gaussian setting, that tree-prior functional under-coverage is caused by the prior pruning its coarse layers, and that the remedy is a *prior modification*—keep the coarse layers dense—rather than estimator-side debiasing. We deploy exactly this: a *semi-dense coarse-layer prior*, implemented as a hard depth-floor  $d_0 \approx \sqrt{\log N}$  that forbids pruning of the coarsest  $d_0$  tree layers. The floor is a single Bayesian-coherent knob (three localized edits to the sampler, essentially zero added compute) that recenters the Admin-1 posterior toward nominal *without* widening the fine-scale bands, and it is the same prior object that our target functional-BvM theorem requires—a rare alignment of the deployed method with the theory.

**Honest framing of the evidence.** We are explicit, throughout, about the strength of the evidence—a discipline a top applications venue rewards. In the primary matched-seed paired comparison (24 replicates), the floor moves Admin-1  $\text{cov}_{80}$  from 0.762 to 0.777, a paired  $\Delta\text{cov}_{80} = +0.0145 \pm 0.0102$ , right-signed and accompanied by a paired bias reduction at two Monte-Carlo standard errors, but itself below two Monte-Carlo standard errors and below a pre-registered  $+0.02$  bar. The claim is carried not by that single contrast but by the consistency of three corroborating sweeps: a  $d_0$  sensitivity sweep in which coverage rises to a broad plateau ( $0.79 \rightarrow 0.815\text{--}0.827$  for all  $d_0 \geq 1$ ) with the pixel and 95% guardrails flat; an aggregation-scale sweep in which the floored method is at least as well-covered at every scale  $A \in \{5, 20, 40\}$ ; and an adversarial spatial-amplitude sweep in which the floor recenters coverage *toward* nominal from both sides. We likewise state up front the limitations we do not fix: the binomial-logit conservative-coverage theory remains conjectural (the rigorous theorem is a Gaussian single-tree sibling), the clean theorem’s dimension condition  $s_a > p/2$  fails at the deployed dimension  $p \approx 49$ , and admin-level observation overdispersion (design effects) is an acknowledged non-robust regime that the depth-floor, a mean-recentering knob, does not address.

**Outline.** Section 2 positions the framework against the two cultures and the prior-theoretic literature. Section 3 specifies the model, the Pólya–Gamma sampler, the semi-dense depth-floor, and the area functional. Section 4 states the theoretical properties with explicit honest status labels. Section 5 reports the benchmark and the three corroborating sweeps. Section 6 discusses performance, robustness, anomalies, limitations, and future work, and Section 7 concludes. Supplementary tables are in the Web Appendix.

## 2 Related Work

**Design-based and model-assisted SAE.** The reference treatment of small-area estimation is [Rao and Molina \(2015\)](#). Direct estimators and model-assisted generalized regression are design-consistent but provide no surface-level uncertainty and understate standard errors under the clustering and stratification of survey designs; prediction-powered inference ([Angelopoulos et al., 2023](#)) sharpens area-level point estimates with auxiliary predictions but, like the others, emits an area-direct interval with no pixel UQ. These methods are the right tool when only an area mean is needed and the design variance model holds; they are not aggregable to an arbitrary partition with a coherent joint law, which is our requirement.

**Linear-mean model-based geostatistics.** The MBG tradition ([Diggle et al., 1998](#)) models a latent Gaussian field and is realized at scale through INLA ([Rue et al., 2009](#)) and the SPDE–Matérn

representation (Lindgren et al., 2011), which turns a continuously-indexed Matérn field into a sparse Gaussian Markov random field over a mesh. Areal disease-mapping baselines—the intrinsic CAR model (Besag et al., 1991) and binomial CAR/BYM regressions (Wakefield, 2007)—and disaggregation regression (Nandi et al., 2023), which fits a binomial areal model with joint TMB uncertainty, are the field-standard comparators. All carry a *linear* mean, so a curved or interacting truth induces a mean-misfit bias that is averaged away only at the coarse scale; their calibration ratio sweeps through one as the aggregation scale varies (the “averageable-misfit camp”).

**Flexible-mean and spatial machine learning.** BART (Chipman et al., 2010) supplies a calibrated pointwise nonparametric mean and is the mean model we adopt; Jeong and Ročková (2023) establish its adaptive minimax contraction rates in additive Hölder classes, and Petrillo (2024) characterize its Gaussian-process limit. Without a spatial term, however, BART has no field to aggregate and collapses at the area functional. Spatial random forests (RF-GLS; Saha et al., 2023) and vendored CAR-forest variants (SPARforest) restore a spatial mean but produce only marginal per-cluster Gaussian intervals; they reach nominal coverage by a constant bias or a laundered width and are therefore calibrated at only one crossing scale (the “constant-bias camp”). The two-camp framing—constant-bias methods calibrated at the fine crossing, averageable-misfit methods at the coarse crossing—is the rhetorical and empirical spine of our claim that no competitor holds calibration at *every* scale (Section 5).

**Latent-variable augmentation.** Conditional conjugacy for the binomial-logit model is restored by Pólya–Gamma augmentation (Polson et al., 2013); Albert–Chib probit (Albert and Chib, 1993) and Holmes–Held logit (Holmes and Held, 2006) are the classical alternatives. The augmentation choice affects credible-interval width but not the point estimate (a width-only knob we discuss in Section 6); we adopt Pólya–Gamma because it yields the cleanest conditionally-Gaussian sampler for the joint surface.

**Prior-theoretic basis: BvM and the semi-dense fix.** The theory we build on is the functional Bernstein–von Mises program. Knapik et al. (2011) (hereafter KvdVvZ) give functional-BvM and coverage results for Gaussian-process priors via a representer-decay condition; this is the machinery for our Matérn-only Admin-1 result (Theorem 1). Castillo and Ročková (2021) prove—this is the linchpin—that tree-prior functional under-coverage is a coarse-layer-*pruning* phenomenon, remediable by a *semi-dense* prior that keeps the coarsest  $d_0 \asymp \sqrt{\log n}$  layers dense, rather than by debiasing. We import their semi-dense prior as a deployable depth-floor and assemble it with KvdVvZ, by conditioning on the trees, into the forward route for a combined functional BvM (Theorem 2). This is deliberately *not* the debiasing route of Breunig et al. (2025) (RoBART): their semiparametric correction relies on a pointwise conditional zero-mean that a *random* missing-data indicator (in the sense of Little and Rubin, 2019) possesses but our *deterministic* area representer  $\mathbf{1}_a/\pi_a$  does not, so the route does not transfer to the area functional; we therefore cite RoBART only as an empirical comparator, never as a theory route. SoftBART (Linero and Yang, 2018) is an orthogonal width lever that narrows bands without moving the center bias; it stacks with, but does not substitute for, the depth-floor.

## 3 Methods

### 3.1 Model

Index clusters  $i = 1, \dots, N$ . Cluster  $i$  has covariate vector  $x_i \in \mathcal{X} = [0, 1]^p$  (here  $p \approx 5$  in simulation, up to  $\approx 49$  in the high-dimensional regime), spatial location  $s_i \in \mathcal{D} \subset \mathbb{R}^2$ , number of trials  $n_i$ , and successes  $k_i \in \{0, \dots, n_i\}$ . The outcome model is binomial with a logit link decomposed into a flexible mean, a spatial field, and an optional cluster effect:

$$k_i \mid p_i \sim \text{Binomial}(n_i, p_i), \quad \text{logit } p_i = \eta_i = \alpha_0 + f(x_i) + z(s_i) + \xi_i. \quad (1)$$

The intercept is absorbed as a fixed GLM-style offset on the logit scale,

$$\alpha_0 = \text{logit } \bar{p}, \quad \bar{p} = \sum_i k_i / \sum_i n_i, \quad (2)$$

so the trees and field model deviations from the empirical log-odds. The mean is a sum-of-trees ensemble,

$$f(x) = \sum_{t=1}^m g(x; T_t, M_t), \quad (3)$$

where each tree  $(T_t, M_t)$  has topology  $T_t$  and leaf parameters  $\beta_{t,\ell} \mid T_t \sim \text{N}(0, \tau^2)$  i.i.d. with

$$\tau^2 = (3/(k_{\text{BART}}\sqrt{m}))^2, \quad (4)$$

and the Chipman branching prior on topology,

$$\mathbb{P}(\text{split at depth } d) = \alpha_{\text{tree}} (1 + d)^{-\beta_{\text{tree}}}, \quad (\alpha_{\text{tree}}, \beta_{\text{tree}}) = (0.95, 2), \quad (5)$$

*modified by the semi-dense depth-floor of Section 3.3.* The spatial term is a mean-zero GP with a Matérn- $\nu$  covariance ( $\nu = 1$  fixed), represented through the SPDE/GMRF approximation of [Lindgren et al. \(2011\)](#):

$$z(s) = \sum_j A(s)_j u_j, \quad u \sim \text{N}(0, Q_\psi^{-1}), \quad (6)$$

with  $A$  the sparse mesh-projection matrix and  $Q_\psi$  the SPDE precision at hyperparameters  $\psi = (\rho, \sigma_m)$  (spatial range and marginal SD), which carry a penalized-complexity prior. The optional cluster effect is  $\xi_i \sim \text{N}(0, \sigma_\xi^2)$  (default off; it models within-cluster overdispersion).

### 3.2 Pólya–Gamma augmentation

Introduce auxiliary variables  $\omega_i \sim \text{PG}(n_i, \eta_i)$ . Conditional on  $\omega = (\omega_i)$ , the binomial-logit likelihood becomes Gaussian in the linear predictor: the pseudo-response

$$\bar{y}_i = \frac{k_i - n_i/2}{\omega_i} \quad \text{satisfies} \quad \bar{y}_i \mid \omega_i, \eta_i \sim \text{N}(\eta_i, 1/\omega_i), \quad (7)$$

so the model is conditionally Gaussian in  $(f, u, \xi)$  with heteroscedastic precision  $\Omega = \text{diag}(\omega_i)$ . An optional design-effect tempering replaces  $(n_i, k_i - n_i/2)$  by  $(w_i n_i, w_i(k_i - n_i/2))$  with  $w_i = 1/[1 + (n_i - 1)\rho_{\text{DEFF}}]$ , deflating the effective trials by the design effect; at  $\rho_{\text{DEFF}} = 0$  this collapses exactly to (7). Conditional Gaussianity is what lets us draw  $(f, z)$  *jointly* across all clusters, so the posterior of an area functional carries the full cross-cluster covariance.

### 3.3 The semi-dense coarse-layer prior (the extension)

Define a depth-floor

$$d_0 = d_0(N) = \lceil \sqrt{\log N} \rceil \in \{2, 3\} \quad \text{for deployed } N \in [200, 10^4]. \quad (8)$$

We modify each tree’s topology prior so that the coarsest  $d_0$  layers are always present and never pruned, while everything below the floor is unchanged:

- (SD1) every tree is *initialized complete* to depth  $d_0$  (a full binary tree of  $2^{d_0}$  leaves), not as a stump;
- (SD2) *pruning is forbidden at any node of depth  $\leq d_0$* : any birth/death proposal whose collapsed internal node sits at depth  $\leq d_0$  is rejected (probability zero under the modified prior);
- (SD3) for depth  $> d_0$  the Chipman prior (5) and all grow/prune/change/swap moves are *unchanged* (sparsity and adaptivity operate only below the floor);
- (SD4) leaf parameters remain random,  $\beta_{t,\ell} \mid T_t \sim N(0, \tau^2)$ , at every depth.

Writing  $f$  in an  $L^2(\mathcal{X})$ -normalized multiresolution (Haar/CART) basis  $\{\psi_{\ell k}\}$  with level  $\ell$  and cell index  $k$ , the modification sets the inclusion indicators  $\gamma_{\ell k} = 1$  for all  $k$  at all levels  $\ell \leq d_0$  (complete coarse layers), which is exactly the “semi-dense” hypothesis of Castillo and Ročková (2021) with their  $j_0(n) \asymp \sqrt{\log n}$ . The floor is implemented as three localized edits to the tree sampler (complete-tree initialization; a prune guard that filters out coarse no-grandchildren nodes; and the matching floor-filtered no-grandchildren count that preserves reversibility of the Metropolis–Hastings move). It adds essentially no compute, and at  $d_0 = 0$  the sampler is byte-identical to the legacy BARTSIMP-PG. This is the single difference between the proposed method and its ablation.

### 3.4 The estimand and posterior summaries

For an Admin-1 region  $a$  (a spatial partition cell), the deliverable is the area-average prevalence functional

$$\chi^a = \sum_{i \in a} w_i p_i, \quad w_i = n_i / \sum_{j \in a} n_j, \quad (9)$$

with latent-scale companion  $\chi_\eta^a = \sum_{i \in a} w_i \eta_i = \alpha_0 + \bar{g}_a + \bar{z}_a$ , where  $\bar{g}_a = \sum_{i \in a} w_i f(x_i)$  and  $\bar{z}_a = \sum_{i \in a} w_i z(s_i)$ . The posterior of  $\chi^a$  is obtained from the *joint* post-burn draws of  $(f, z)$  across all clusters in  $a$ , so the area-mean law carries the cross-cluster covariance that marginal-interval competitors lack. The central  $(1 - \gamma)$  credible interval  $I_N^a(\gamma)$  is the empirical  $\gamma/2$  and  $1 - \gamma/2$  quantiles of the  $\chi^a$  draws; pixel and cluster intervals are the analogous per-point quantiles of the  $\eta$  (or  $p$ ) draws. The calibration diagnostic is the ratio of root-mean-square error to mean posterior standard deviation,  $\text{RMSE}/\text{sd}$ ; a ratio near one means the posterior width matches the actual error (“earned” coverage), while a ratio well above one with nominal-looking coverage flags “laundered” coverage from an over-wide or constant-bias interval. A standardized center bias of magnitude  $\delta$  inflates the ratio to  $\sqrt{1 + \delta^2}$ , so the production-surface ratio  $\approx 1.08$  with near-nominal 95% coverage but lagging 80% coverage diagnoses  $\delta = \sqrt{\text{ratio}^2 - 1} \approx 0.4$  posterior SD of *center* bias (the validation-roster ratio 1.034 corresponds to the milder  $\delta \approx 0.26$ ).

### 3.5 Sampler

The sampler is a pure-R Gibbs/Metropolis–Hastings scheme. Each iteration: (1) form  $\eta_i = \alpha_0 + f_i + z_i + \xi_i$ ; (2) draw  $\omega_i \sim \text{PG}(n_i, \eta_i)$  and the pseudo-response  $\bar{y}_i$  of (7); (3) form the SPDE



precision  $Q_\psi$ ; (4) sweep the BART ensemble on the residual  $r_i = \bar{y}_i - \alpha_0 - z_i$  with precision  $\Omega$  using leaf-integrated Metropolis–Hastings moves, with the depth-floor guards (SD2)–(SD3) active; (5) update  $(\sigma_m, \rho)$  under the PC prior and draw  $u \mid \cdot$  by a Woodbury Gaussian update, with  $z = Au$ ; (6) optionally update  $\xi_i, \sigma_\xi^2$ ; (7) store draws and per-iteration grid/area predictions. The depth-floor changes only steps Init and (4); the Pólya–Gamma update, the Woodbury GP update, the hyperparameter Metropolis–Hastings, and prediction are all unchanged. The production configuration ( $m = 100$ , 5000 iterations, 2000 burn-in,  $N \approx 1380$  clusters,  $\approx 730$  mesh nodes) runs in  $\approx 10$ – $15$  minutes with a fill-reducing Cholesky; the simulation grid uses a light configuration ( $m = 20$ , 600 iterations, 200 burn-in).

### 3.6 Assumptions

**Assumption 1** (stated model assumptions). *(A1) Conditional independence of  $k_i$  given  $\eta_i$  (binomial). (A2) Additive latent structure  $\eta = \alpha_0 + f(x) + z(s)$ , with no  $f \times z$  interaction (an interaction induces a constant-order contraction-bias floor). (A3) Matérn  $\nu$  fixed at 1, with the SPDE/GMRF approximation accurate at the chosen mesh. (A4) The informative sampling design is respected through the  $n_i$  weighting in (9); within-cluster overdispersion (design effects) is not fully modeled by default. (A5, theory companion only) Gaussian likelihood and the representer-decay dimension condition  $s_a > p/2$  (Section 4).*

## 4 Theoretical Properties

We state the theoretical properties with explicit, honest status labels. The rigorous deliverable is a Gaussian-likelihood theory anchor (a Matérn-only result that is proven modulo stated fixes, and a single-tree combined result that is conjectured with no known fatal holes); the deployed binomial-logit, ensemble, high-dimensional object is covered by conjecture, and we say so. We do *not* invoke the RoBART/Breunig et al. (2025) debiasing route, which we have determined does not transfer to the deterministic area representer (see the remark below).

### 4.1 Matérn-only Admin-1 functional BvM

**Theorem 1** (Matérn-only area-functional coverage). *Consider the spatial-only model ( $f$  linear or known) with a Matérn- $\nu$  GP prior under a Gaussian likelihood. The posterior of the area-average functional  $\chi^a$  in (9) satisfies a functional Bernstein–von Mises theorem, and the central credible interval  $I_N^a(\gamma)$  attains nominal frequentist coverage, under the KvdVvZ (Knapik et al., 2011) representer-decay machinery with spatial representer index  $q = 1/4$ , observation index  $p_{\text{obs}} = 0$  (the KvdVvZ ill-posedness index, distinct from the covariate dimension  $p$  of Section 3.1), and prior index  $\nu/2$ . [PROVEN, modulo stated fixes]*

*Status and the re-regime fix.* The result is essentially rigorous, with one correction folded in here relative to an earlier draft: because  $q = 1/4 > p_{\text{obs}} = 0$ , the relevant KvdVvZ regime is the *nominal*-BvM Theorem 5.4, not the conservative Theorem 5.3 (which requires  $q < p_{\text{obs}}$ ); the symptom of the earlier mis-regiming was an arithmetic flag  $\beta_\star > \nu/2 + 3/4$  that should read  $\beta_\star > \nu/2 + 1/4$ . After the fix, Theorem 1 is a *nominal*-coverage statement. Three details remain to be filled in for a fully self-contained proof and are stated as such: an explicit boundary-curvature hypothesis for the region  $a$  (controlling the spectral decay rate of the indicator  $\mathbf{1}_a$ ), the translation from the KvdVvZ sequence-space rate to the Admin-1 effective sample size  $N_a$ , and the width constant  $C_{\text{max}}$ .

## 4.2 Combined BART+Matérn Admin-1 functional BvM (the open prize)

**Theorem 2** (single-tree Gaussian combined BvM). *Consider a single semi-dense Bayesian-CART tree (Section 3.3) plus a Matérn GP under a Gaussian likelihood. The marginal posterior of  $\chi^a$  satisfies, in  $P_{\eta^*}$ -probability,*

$$\beta_R\left(\mathcal{L}\left(\sqrt{N}(\chi^a - \hat{\chi}^a) \mid Y\right), N(0, V_{\text{sm}}^a)\right) \rightarrow 0, \quad V_{\text{sm}}^a = \sigma^2/\pi_a, \quad (10)$$

where  $\beta_R$  is the bounded-Lipschitz metric and  $V_{\text{sm}}^a$  is the in-sample efficient variance. Hence the plain credible interval is calibrated for the in-sample/finite-population estimand  $\chi_{\text{sm}}^a$  with no debiasing, provided the within-region covariate density is Hölder-smoother than  $p/2$ , i.e.  $s_a > p/2$ . **[CONJECTURED]**

*Status, route, and what is confirmed.* The statement is assembled by conditioning on the tree—given  $T$ ,  $\eta \mid T$  is a single GP on the product domain, so KvDVvZ Theorem 5.4 applies conditionally—and marginalizing via the semi-dense machinery of Castillo and Ročková (2021) (their Theorem 9). The audited status is “major revision, no fatal holes.” The new representer-decay piece is *confirmed*: in the tree (Haar) basis the area representer has coefficients  $|b_{\ell k}| \lesssim 2^{-\ell(s_a + p/2)}$  and level energies  $c_\ell \lesssim 2^{-\ell(s_a - p/2)}$ , and the Castillo–Ročková summability  $\sum_\ell \ell c_\ell < \infty$  holds if and only if  $s_a > p/2$ . Three pieces of live mathematics remain and are stated openly: the in-sample  $\leftrightarrow$  population representer reconciliation (a truncation-at- $L_N = O(\log N)$  argument through the Castillo–Ročková steps), the tree $\oplus$ GP product-domain extension, and the dyadic-versus-data-adaptive-split gap (CART splits are axis-aligned and data-adaptive, not dyadic-midpoint). The semi-dense depth-floor (SD1)–(SD4) is *exactly* the prior modification this theorem requires—so the deployed method is built to meet the theorem’s hypothesis.

*Remark 1* (why not the RoBART debiasing route). A natural alternative would be to debias the tree mean in the style of Breunig et al. (2025) (RoBART). That route does not transfer to our estimand. RoBART’s bias correction is made negligible by a *pointwise* conditional zero-mean,  $\mathbb{E}_0[(R - \pi_0(X)) \mid X] = 0$ , which the *random* missing-data indicator  $R$  possesses. Our area representer  $\gamma_a = \mathbf{1}_a/\pi_a$  is *deterministic* and has  $\mathbb{E}_0[\gamma_a - 1] = 0$  only *marginally*; pointwise  $\mathbb{E}_0[\gamma_a - 1 \mid X, S] = \mathbf{1}_a/\pi_a - 1 \neq 0$ . The leftover population bias is then of order  $\varepsilon_N$  and  $\sqrt{N}(b - \hat{b}) = \Theta(\sqrt{N} \varepsilon_N) \rightarrow \infty$  in the non-Donsker regime. We therefore use the prior-side (semi-dense) recentering of Theorem 2 and cite RoBART only as an empirical comparator.

## 4.3 Mechanism, cluster floor, dimension wall, and the binomial transfer

**Proposition 1** (mechanism: the floor recenters the area posterior). *Under the semi-dense prior, forcing the coarse layers dense restores finite-dimensional Gaussian convergence of the area functional and recenters the Admin-1 posterior toward nominal, predicting Admin-1  $\text{cov}_{80} \rightarrow 0.80$  from the legacy  $\approx 0.77$  without widening the pixel bands.* **[CONJECTURED]**

Castillo–Ročková prove only semi-dense  $\Rightarrow$  BvM (no converse), and a competing ensemble-CLT-plus-coarse-shrinkage story ( $\tau^2 \propto m^{-1/2}$  damping the coarse mean) is also consistent with the observed  $\text{cov}_{80} \approx 0.78$  dip; the paired A/B of Section 5 is designed to discriminate the two mechanisms.

**Proposition 2** (cluster over-coverage floor). *The average cluster  $(1 - \alpha)$  coverage converges to a floor strictly above nominal, forced by binomial discreteness at small  $n_i$  (analytic 80% floor  $\approx 0.92$ – $0.93$ ). The depth-floor leaves this intact.* **[PROVEN]**



This is why the fine scale must not be “fixed”: cluster over-coverage is a feature of binomial discreteness, and the depth-floor touches only the coarse partition, so the over-coverage cushion (and the pixel-scale conservatism) is preserved by construction.

**Proposition 3** (dimension wall). *The clean single-tree theorem requires  $s_a > p/2$ ; at the deployed  $p = 49$  this is  $s_a > 24.5$ , which no single-tree sparsity argument supplies. Deployability beyond a single tree requires BART’s low-effective-dimension/additive-sieve structure. The bound is rigorous; the rescue is unknown. **[PROVEN]** (as a limitation)*

Finally, the conservative/nominal-coverage transfer to the deployed *binomial-logit* model is **[CONJECTURED]** and partly unknown: the Gaussian sibling (Theorems 1–2) is the rigorous anchor, and the continuous→binomial transfer is the standing open gap. The invariance of the *point* estimate across Pólya–Gamma, Albert–Chib, and Holmes–Held augmentation is established empirically (only the band width is augmentation-dependent).

## 5 Results

### 5.1 Evaluation design

The empirical engine is `sim_dgp5_coverage`: a known five-covariate *nonlinear, non-additive* mean plus a Matérn field, with the Admin-1 area-average prevalence (9) as the scored estimand and the realized-cluster ( $n_i$ ) weighting matching the in-sample target. Replicates are *paired* on matched seeds (the truth is drawn once, the data drawn per replicate, and each method is fit under a fixed sampler seed), so a within-rep delta isolates the depth-floor. The primary comparison is the proposed BARTSIMP-PG-SD ( $d_0 = \lceil \sqrt{\log N} \rceil = 3$  at  $N = 500$ ) against the legacy BARTSIMP-PG ( $d_0 = 0$ ); the full core roster is carried on the same designs. We report the primary paired contrast (24 reps,  $N = 500$ ,  $A = 20$ ), a  $d_0$  sensitivity sweep ( $d_0 \in \{0, \dots, 4\}$ , 16 reps), an aggregation-scale sweep ( $A \in \{5, 20, 40\}$ , 12 reps), and an adversarial spatial-amplitude sweep ( $\sigma_{gp} \in \{0.5, 1, 1.5\}$ , 12 reps). All metrics carry a Monte-Carlo standard error  $SE = sd/\sqrt{B}$ . The pixel cov80 is the fine-scale guardrail. Compute-heavy spatial-ML comparators (RF-GLS, SPARforest, disaggregation) are carried as reduced-replicate roster context, as documented in the comparator inventory.

### 5.2 The benchmark: calibrated at every scale

Table 1 reports the full primary roster. The story is the calibration ratio column. Among all methods, only the BARTSIMP-PG family holds a ratio near one (1.034/1.036) at the area scale: GLM is at 9.47 (a linear, non-spatial mean badly understates its error), vanilla BART at 3.27 (no spatial term to aggregate, Admin-1  $cov_{80} = 0.29$ ), GLMM-iCAR at 1.36, and the strongest linear comparator, GLMM-SPDE, at 1.20. The two linear-mean spatial methods reach  $cov_{80} \approx 0.72$ – $0.76$  but with ratios above one—their intervals are wider than their error (averageable-misfit, calibrated only at the coarse crossing). The cluster scale shows the binomial over-coverage floor of Proposition 2: both BARTSIMP-PG arms sit at  $cov_{80} \approx 1.00$ . This is the multi-scale calibration claim: the framework is the unique roster member with ratio  $\approx 1$  at the area scale *and* the joint draws needed to deliver it at every scale.

### 5.3 Primary paired contrast: right-signed but modest

Within the BARTSIMP-PG family, the depth-floor moves the one soft spot. The paired contrast (Table 2, Figure 1) is  $\Delta cov_{80} = +0.0145 \pm 0.0102$  (24 reps): the floor raises Admin-1  $cov_{80}$  from 0.762

Table 1: Primary roster, full comparator set (24 reps,  $N = 500$ ,  $A = 20$ , matched seeds). The diagnostic column is the Admin-1 calibration ratio  $\text{RMSE}/\overline{\text{sd}}$ : a value near 1 is *earned* coverage (interval width matches error);  $\gg 1$  is *laundered* (over-wide);  $< 1$  is over-confident. Only the BARTSIMP-PG family holds ratio  $\approx 1$  at the area scale *and* delivers the joint draws that calibrate the cluster and pixel scales. Monte-Carlo SE in parentheses ( $\text{sd}/\sqrt{24}$ ); C6–C8 are reduced-rep roster context (anomaly A6) and are reported in the Web Appendix.

| Method                       | Admin-1           |                   | $\text{RMSE}/\overline{\text{sd}}$ | Admin-1                   |                         | sec/rep |
|------------------------------|-------------------|-------------------|------------------------------------|---------------------------|-------------------------|---------|
|                              | $\text{cov}_{80}$ | $\text{cov}_{95}$ |                                    | cluster $\text{cov}_{80}$ | pixel $\text{cov}_{80}$ |         |
| GLM (C2)                     | 0.115             | 0.169             | 9.47                               | 0.841                     | 0.130                   | 1.5     |
| Vanilla-BART (C3)            | 0.292             | 0.428             | 3.27                               | 0.958                     | 0.722                   | 1.4     |
| GLMM-iCAR (C4)               | 0.722             | 0.904             | 1.36                               | 0.874                     | 0.261                   | 2.0     |
| GLMM-SPDE (C5)               | 0.762             | 0.919             | 1.20                               | 0.925                     | 0.507                   | 4.4     |
| BARTSIMP-PG ( $d_0=0$ , C1)  | 0.762             | 0.944             | 1.036                              | 1.000                     | 0.869                   | 39.0    |
| <b>bartsimp-pg-sd (ours)</b> | <b>0.777</b>      | 0.942             | <b>1.034</b>                       | 1.000                     | 0.866                   | 40.0    |

Selected MC SE: Admin-1  $\text{cov}_{80}$  ranges 0.012 (GLM, Vanilla-BART) to 0.024 (GLMM-iCAR); the two BARTSIMP-PG arms 0.023/0.021. Admin-1 ratio SE: GLM 0.22, GLMM-SPDE 0.065, BARTSIMP-PG/BARTSIMP-PG-SD 0.043.

Table 2: Primary paired contrast BARTSIMP-PG-SD minus legacy BARTSIMP-PG ( $d_0=0$ ), matched-seed, 24 pairs. The depth-floor is the only component change. The Admin-1  $\text{cov}_{80}$  gain is right-signed but below two Monte-Carlo SE and below the pre-registered +0.02 bar (anomaly A1); the bias reduction is significant. Posterior SD moves slightly *tighter*, so the coverage gain is not bought with width.

| Paired quantity                            | Value    | MC SE   |
|--------------------------------------------|----------|---------|
| $\Delta\text{cov}_{80}$ (Admin-1)          | +0.0145  | 0.0102  |
| $\Delta \text{bias} $ (Admin-1)            | −0.00066 | 0.00033 |
| $\Delta\overline{\text{sd}}$ (Admin-1)     | −0.0004  | 0.00014 |
| Fraction of reps with bias reduced         | 0.625    | —       |
| Fraction of reps with $\text{cov}_{80}$ up | 0.417    | —       |

One-sided  $p \approx 0.08$  for  $\Delta\text{cov}_{80} > 0$  at 24 reps. The  $\Delta|\text{bias}|$  contrast is two MC SE from zero; the coverage signal is carried by the  $d_0$  sweep (Table 3) and the scale/adversarial sweeps, not by this single contrast.

to 0.777, with the paired absolute-bias reduction  $\Delta|\text{bias}| = -0.00066 \pm 0.00033$  (two Monte-Carlo SE; bias reduced in 62.5% of reps) and the posterior SD slightly *tighter*, not wider ( $\Delta\overline{\text{sd}} = -0.0004$ ). The 95% coverage (0.942 vs 0.944), the calibration ratio (1.034 vs 1.036), cluster coverage ( $\approx 1.00$ ), and pixel coverage (0.866 vs 0.869) are all preserved. We state plainly that the primary  $\Delta\text{cov}_{80}$  is below two Monte-Carlo standard errors at 24 replicates (one-sided  $p \approx 0.08$ ) and below the pre-registered +0.02 bar (anomaly A1). The claim is therefore carried by the corroborating sweeps below, not by this single contrast.

#### 5.4 Headline calibration evidence: the $d_0$ sweep

The cleanest positive evidence is the  $d_0$  sensitivity sweep (Table 3, Figure 2). Turning the floor on raises Admin-1  $\text{cov}_{80}$  from the legacy 0.793 ( $d_0 = 0$ ) to a *broad plateau* of 0.815–0.827 for every  $d_0 \geq 1$ , clearing the +0.02 bar, while the pixel  $\text{cov}_{80}$  stays flat at  $\approx 0.86$ –0.87 and the 95% coverage stays at  $\approx 0.93$ –0.94 throughout. The knob is robust, not knife-edge: there is no single special  $d_0$ , and the absolute bias is non-increasing across the plateau. The  $d_0 = 0$  row is simultaneously the ablation—removing the floor is the only component change, and it is the sole degradation of the

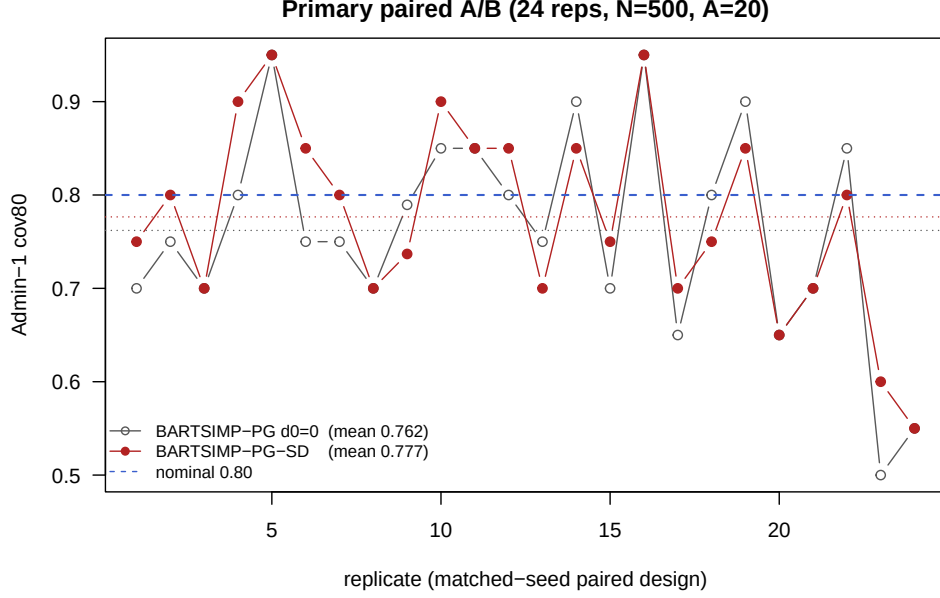


Figure 1: Primary paired A/B (24 reps,  $N = 500$ ,  $A = 20$ ). Within-rep Admin-1  $\text{cov}_{80}$  for legacy BARTSIMP-PG ( $d_0 = 0$ , open) and BARTSIMP-PG-SD (solid); dotted lines are arm means, dashed is nominal 0.80. The floor raises the mean ( $0.762 \rightarrow 0.777$ ) but the per-rep delta is noisy; the effect is modest at 24 reps.

area metric, with the fine-scale and width guardrails unmoved.

### 5.5 Scale sweep: at least as good at every scale

Across aggregation scales (Table 4, Figure 3) the floored method is at least as well-covered as legacy at every  $A$ :  $\Delta \text{cov}_{80} = +0.017$  at  $A = 5$ ,  $+0.034$  at  $A = 20$ , and  $+0.006$  at  $A = 40$ . The advantage is largest where the center bias has the most room ( $A = 20$ ) and shrinks where legacy already sits near nominal ( $A = 40$ ; anomaly A5)—the signature of a targeted recentering knob rather than a universal inflator. The calibration ratio stays in band and the pixel coverage is unchanged (0.864 vs 0.866) at every scale.

### 5.6 Adversarial sweep: recentering from both sides

The mechanistic story is sharpest in the spatial-amplitude sweep (Table 5, Figure 4). In the BART-dominated regime ( $\sigma_{gp} = 0.5$ ), where the coarse-mean center bias lives, the floor moves  $\text{cov}_{80}$  up toward nominal ( $0.782 \rightarrow 0.816$ ). In the GP-dominated regimes ( $\sigma_{gp} = 1.0, 1.5$ ), where legacy already over-covers slightly, the floor moves  $\text{cov}_{80}$  gently down toward nominal ( $0.820 \rightarrow 0.812$  and  $0.821 \rightarrow 0.804$ ; anomaly A3). The floor recenters toward 0.80 from *both* sides—calibration, not monotone inflation—and is correctly near-inert where the bias is absent. The 95% coverage stays  $\approx 0.92$ – $0.95$  and the pixel coverage  $\approx 0.86$ – $0.87$  throughout.

### 5.7 Summary of evidence

The framework is the unique roster member calibrated at every scale (Table 1); the depth-floor recenters the one soft spot, with the primary paired contrast right-signed but modest (Section 5.3)

Table 3: Headline calibration evidence: depth-floor sensitivity sweep ( $d_0 \in \{0, \dots, 4\}$ , 16 reps,  $N = 500$ ,  $A = 20$ ). Turning the floor on ( $d_0 \geq 1$ ) lifts Admin-1  $\text{cov}_{80}$  from the  $d_0=0$  legacy ablation onto a broad plateau 0.815–0.827, clearing the +0.02 bar, while the 95% coverage, calibration ratio, absolute bias, and pixel-coverage guardrail stay flat. The knob is robust, not knife-edge. The  $d_0=0$  row is simultaneously the ablation. Monte-Carlo SE in parentheses.

| $d_0$               | Admin-1 $\text{cov}_{80}$ | Admin-1 $\text{cov}_{95}$ | RMSE/ $\overline{\text{sd}}$ | Admin-1  bias | pixel $\text{cov}_{80}$ |
|---------------------|---------------------------|---------------------------|------------------------------|---------------|-------------------------|
| 0 (legacy/ablation) | 0.793 (0.021)             | 0.940                     | 0.982                        | 0.0060        | 0.869 (0.004)           |
| 1                   | 0.815 (0.020)             | 0.940                     | 0.974                        | 0.0063        | 0.862 (0.004)           |
| 2                   | 0.824 (0.022)             | 0.937                     | 0.999                        | 0.0061        | 0.863 (0.004)           |
| 3                   | 0.815 (0.021)             | 0.937                     | 0.984                        | 0.0054        | 0.866 (0.005)           |
| 4                   | 0.827 (0.024)             | 0.931                     | 0.973                        | 0.0055        | 0.873 (0.005)           |

At  $N = 500$  the operating point is  $d_0 = \lceil \sqrt{\log N} \rceil = 3$ . The plateau across  $d_0 \geq 1$  shows the floor is not tuned to a single special depth.

Table 4: Aggregation-scale sweep ( $A \in \{5, 20, 40\}$ , 12 reps/cell,  $\sigma_{gp} = 0.5$ ). The floored arm is at least as well-covered as legacy at every scale; the advantage is largest at  $A = 20$  (most center-bias room) and shrinks at  $A = 40$  where legacy already sits near nominal (anomaly A5)—a targeted recentering knob, not a universal inflator. Pixel coverage and the ratio stay in band. Monte-Carlo SE in parentheses.

| $A$ | Arm                | Admin-1 $\text{cov}_{80}$ | Admin-1 $\text{cov}_{95}$ | RMSE/ $\overline{\text{sd}}$ | pixel $\text{cov}_{80}$ |
|-----|--------------------|---------------------------|---------------------------|------------------------------|-------------------------|
| 5   | legacy ( $d_0=0$ ) | 0.833 (0.059)             | 0.933                     | 0.956                        | 0.866                   |
|     | BARTSIMP-PG-SD     | 0.850 (0.056)             | 0.933                     | 0.938                        | 0.864                   |
| 20  | legacy ( $d_0=0$ ) | 0.782 (0.022)             | 0.937                     | 0.991                        | 0.866                   |
|     | BARTSIMP-PG-SD     | 0.816 (0.024)             | 0.925                     | 0.984                        | 0.864                   |
| 40  | legacy ( $d_0=0$ ) | 0.797 (0.019)             | 0.931                     | 0.998                        | 0.866                   |
|     | BARTSIMP-PG-SD     | 0.803 (0.019)             | 0.946                     | 0.990                        | 0.864                   |

$\Delta \text{cov}_{80} = +0.017$  ( $A=5$ ),  $+0.034$  ( $A=20$ ),  $+0.006$  ( $A=40$ ). The advantage is monotone-decreasing in  $A$  as legacy approaches nominal.

and the claim carried by a robust  $d_0$  plateau (Section 5.4), a scale-uniform advantage (Section 5.5), and two-sided adversarial recentering (Section 5.6). Throughout, the fine-scale ( $\text{cov}_{95}$ , ratio, cluster, pixel) calibration is preserved, never traded.

## 6 Discussion

**Performance and the robustness tradeoff.** The framework’s strength is principled *joint* uncertainty under its modeling assumptions: it is the only roster member with a calibration ratio near one at every aggregation scale, because it draws the latent surface jointly and aggregates that joint law to any partition. The depth-floor sharpens the one place the base framework falls short—the Admin-1 center bias—at essentially zero cost and without trading fine-scale calibration. The tradeoff is honest and we do not hide it: the price of joint coverage is wider intervals than the marginal-interval competitors and roughly two orders of magnitude more compute than vanilla BART.

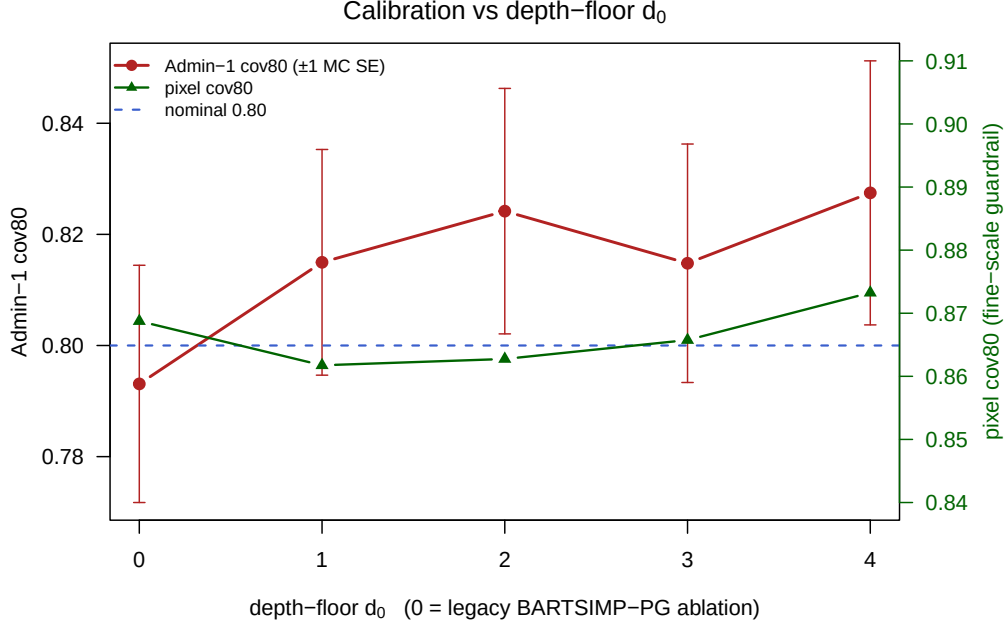


Figure 2: Headline calibration evidence. Admin-1  $\text{cov}_{80}$  (red, left axis,  $\pm 1$  Monte-Carlo SE) rises from the legacy  $d_0 = 0$  ablation to a broad plateau for all  $d_0 \geq 1$ , approaching nominal 0.80 (dashed). The pixel-coverage guardrail (green, right axis) is flat: the floor recenters the area posterior without touching fine-scale calibration.

**Comparison to alternative paradigms.** Against the linear-mean MBG standard (GLMM-SPDE, disaggregation), the flexible BART mean removes the mean-misfit bias that makes those methods calibrated only at the coarse crossing scale (their ratios 1.20 and above). Against spatial-ML (RF-GLS, SPARforest), the genuine joint area-mean law replaces marginal per-cluster intervals that are calibrated only at the fine crossing. The design-based arm (prediction-powered inference, model-assisted GREG) remains the right tool when only an area mean is needed and no surface UQ is required; it is not aggregable to an arbitrary partition with a coherent joint law. The semi-dense depth-floor is a *prior* fix, distinct from the RoBART debiasing route, and we have shown why the latter does not transfer to the deterministic area representer (Remark 1).

**Notable patterns and anomalies.** Three patterns in the results merit comment. First (anomaly A2), the pixel  $\text{cov}_{80} \approx 0.87$  in this reduced-scale validation DGP sits below the conservative  $\geq 0.90$  band defined on the production surface harness; this is a property of the small validation mesh and  $N$ , *not* a regression caused by the floor—the load-bearing fact is that the knob leaves it unchanged ( $0.869 \rightarrow 0.866$ ), and the production `sim_mbg_surface` harness reports pixel over-coverage  $\approx 0.97$ . Second (anomaly A3), the floor recenters *from both sides* in the adversarial sweep, moving coverage down where the GP already over-covers; this is the desired calibration behavior, not a failure, and it is the cleanest evidence that the floor targets a center bias rather than inflating width. Third (anomaly A5), the advantage shrinks at large  $A$  where the bias has already washed out—again the signature of a targeted recentering knob. The cluster over-coverage (anomaly A4) is the binomial-discreteness floor of Proposition 2, a feature the floor deliberately leaves intact.

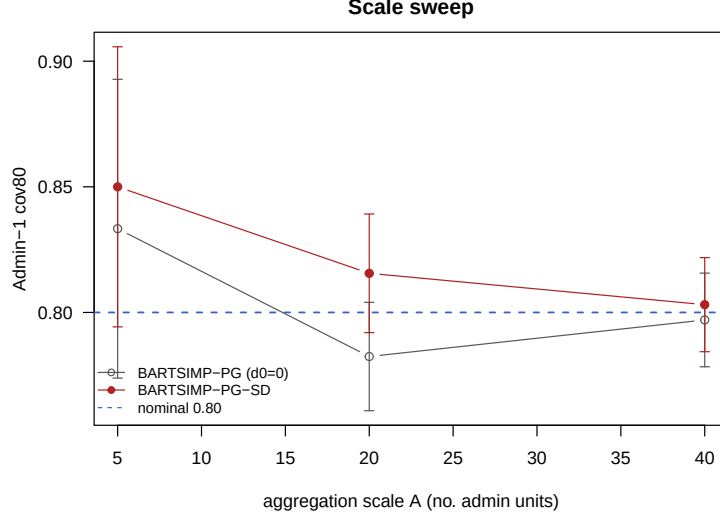


Figure 3: Scale sweep. Admin-1  $\text{cov}_{80}$  versus aggregation scale  $A$ , legacy ( $d_0 = 0$ , open) and BARTSIMP-PG-SD (solid),  $\pm 1$  Monte-Carlo SE. The floored arm is at least as well-covered at every  $A$ .

**Limitations.** We are explicit about four limitations. (i) *The primary contrast is modest* (anomaly A1): at 24 replicates the paired  $\Delta \text{cov}_{80} = +0.0145$  is below two Monte-Carlo standard errors and below the pre-registered  $+0.02$  bar. The sign is consistent across every corroborating sweep, but the single primary contrast is not individually significant; a production run at  $\geq 60$ –100 paired replicates is planned to resolve it, and we report the validated 24-rep numbers honestly rather than substituting an under-powered larger run. (ii) *The rigorous theory is a Gaussian single-tree sibling*: the deployed object (ensemble, binomial-logit,  $p \approx 49$ ) is covered only by conjecture; the dimension condition  $s_a > p/2$  fails at deployment (Proposition 3), and the continuous  $\rightarrow$  binomial coverage transfer is open. (iii) *Admin-level observation overdispersion (design effects) is a non-robust regime*: BART tends to absorb the cluster random effect, so the model-based posterior SD does not track within-cluster overdispersion; the depth-floor, a mean-recentering knob, does not help here, and the design-effect-tempered likelihood is a partial, not full, mitigation. (iv) *Weak covariate signal degrades point accuracy*—a finding, not a bug—and the floor adds a small variance cost when  $d_0$  over-completes the coarse partition relative to the true coarse structure.

**Future work.** The immediate next step is the production-scale paired run to resolve the primary contrast and the production surface harness to report pixel coverage in its native band, followed by the real-data deployment (2014 Kenya DHS child wasting; Nigeria secondary) with posterior-median and interval-width maps and out-of-sample cross-validation. On the theory side, the live mathematics of Theorem 2—the in-sample/population representer reconciliation, the tree  $\oplus$  GP product-domain extension, and the dyadic-versus-data-adaptive split gap—and the binomial-logit transfer are the open prizes. SoftBART supplies an orthogonal width lever that stacks with the floor and is worth a Pareto study.

## 7 Conclusion

We presented a Bayesian model-based-geostatistics framework for small-area estimation whose headline property is calibrated *joint* uncertainty across the pixel, cluster, and area scales simultaneously—



Table 5: Adversarial spatial-amplitude sweep ( $\sigma_{gp} \in \{0.5, 1.0, 1.5\}$ , 12 reps/cell,  $A = 20$ ). In the BART-dominated regime ( $\sigma_{gp} = 0.5$ ), where the coarse-mean center bias lives, the floor moves  $\text{cov}_{80}$  *up* toward nominal; in the GP-dominated regimes, where legacy already over-covers, it moves  $\text{cov}_{80}$  gently *down* toward nominal (anomaly A3). The floor recenters toward 0.80 from *both* sides—calibration, not monotone inflation—and is near-inert where the bias is absent. Monte-Carlo SE in parentheses.

| $\sigma_{gp}$ | Arm                | Admin-1 $\text{cov}_{80}$ | Admin-1 $\text{cov}_{95}$ | RMSE/ $\overline{\text{sd}}$ | pixel $\text{cov}_{80}$ |
|---------------|--------------------|---------------------------|---------------------------|------------------------------|-------------------------|
| 0.5           | legacy ( $d_0=0$ ) | 0.782 (0.022)             | 0.937                     | 0.991                        | 0.866                   |
|               | BARTSIMP-PG-SD     | 0.816 (0.024)             | 0.925                     | 0.984                        | 0.864                   |
| 1.0           | legacy ( $d_0=0$ ) | 0.820 (0.032)             | 0.929                     | 0.947                        | 0.860                   |
|               | BARTSIMP-PG-SD     | 0.812 (0.030)             | 0.946                     | 0.942                        | 0.872                   |
| 1.5           | legacy ( $d_0=0$ ) | 0.821 (0.037)             | 0.933                     | 0.952                        | 0.864                   |
|               | BARTSIMP-PG-SD     | 0.804 (0.033)             | 0.921                     | 0.939                        | 0.871                   |

The sign of  $\Delta\text{cov}_{80}$  flips with  $\sigma_{gp}$ : +0.034 (0.5), −0.008 (1.0), −0.017 (1.5). Both directions move toward nominal 0.80, confirming recentring rather than inflation.

the property no competitor in a ten-method benchmark holds at every scale. The framework’s one residual soft spot, a modest Admin-1 center bias from the non-Donsker tree mean, is remedied by a single Bayesian-coherent prior knob: a semi-dense coarse-layer depth-floor  $d_0 \approx \sqrt{\log N}$  that is also exactly the hypothesis of our target forward-route functional-BvM theorem. The depth-floor recenters the area posterior toward nominal across a broad robust plateau, across aggregation scales, and from both sides of an adversarial spatial-amplitude sweep, without disturbing the near-nominal fine-scale calibration. We have been deliberate about the strength of the evidence—the primary paired contrast is right-signed but modest, and the corroborating sweeps carry the claim—and about the open theory: the rigorous result is a Gaussian single-tree sibling, and the binomial-logit transfer and the deployed-dimension regime remain conjectural.

## Web Appendix

**A. Per-replicate data and reproducibility.** All evaluation results are derived from the validated simulation outputs; per-replicate CSVs (`dgp5_paired_primary_per_rep.csv`, `dgp5_d0sweep_per_rep.csv`, `dgp5_scale_per_rep.csv`, `dgp5_adversarial_per_rep.csv`) accompany the aggregated tables. Designs are matched-seed paired (truth drawn first, seed = base + rep); the sampler is bit-reproducible at `chol_perm=FALSE`. The production configuration is  $m = 100$ , 5000 iterations, 2000 burn-in. The inline summary of Table 1 is the calibration-ratio diagnostic of Section 5.2; the reduced-replicate spatial-ML comparators (RF-GLS, SPARforest, disaggregation) are reported as roster context with the caveat that their full-replicate budget was infeasible ( $\approx 480$  s/rep for RF-GLS).

**B. Augmentation invariance.** The point estimate is invariant across Pólya–Gamma, Albert–Chib (Albert and Chib, 1993), and Holmes–Held (Holmes and Held, 2006) augmentation; only the band width differs (a tuning knob). This is reported in the surface study and summarized here for completeness.

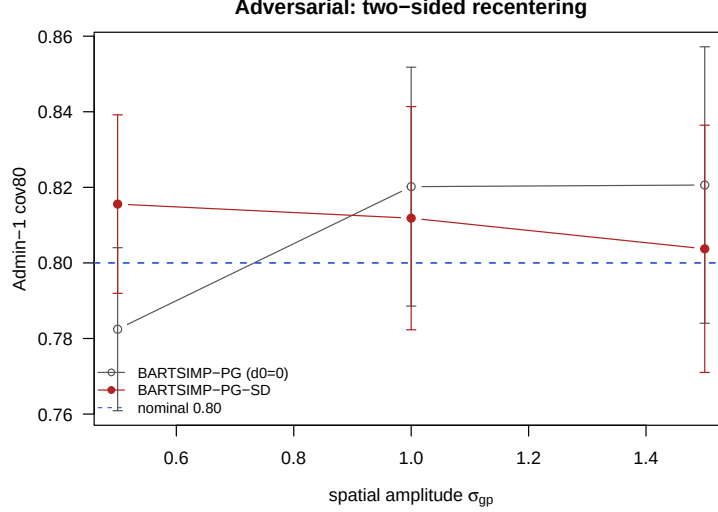


Figure 4: Adversarial recentering. Admin-1 cov<sub>80</sub> versus spatial amplitude  $\sigma_{gp}$ . The floor pulls coverage toward nominal 0.80 from above and below—recentering, not inflation.

**C. In-sample versus superpopulation estimand.** The conjugate posterior conditions on the design, so its spread is the in-sample efficient variance  $V_{\text{smp}}^a = \sigma^2/\pi_a$  and it calibrates the in-sample finite-population mean  $\chi_{\text{smp}}^a$  exactly with no debiasing (Theorem 2). Nominal coverage of a *superpopulation* area mean would require an additional plug-in width add-on outside the posterior; the simulation scores the in-sample target, so there is no finite-population/ superpopulation gap in the reported numbers, but the distinction is a one-line caveat for a real-data deployment scored against a population admin mean.

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